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Divakar Yadav

MS, Computer Science

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Summary

Passionate Software Engineer with an MS in Computer Science and **4+ years of experience** developing scalable web applications, microservices-based backend systems, and GPU-accelerated ML pipelines. Skilled in building high-performance **CUDA kernels**, profiling GPU workloads using **Nsight Systems/Compute**, and integrating optimized inference workflows with **PyTorch** and **TensorRT**. Experienced in both startup and enterprise environments, collaborating across teams to deliver production-ready systems. Actively working on **H100/A100 GPU tuning**, **multi-GPU scheduling**, and **memory-efficient model deployment** for real-time inference at scale.

Education

M.S, Computer Science, University of Wisconsin, Milwaukee **Sep 2023 – May 2025**

B.Tech, Feroze Gandhi Institute of Engineering & Technology, Raebareli **July 2014 – May 2018**

Relevant Courses

Concurrent Programming, Machine Learning, Software Defined Networks , Text Retrieval from Documents, Natural Language Processing, Databases, Operating Systems, Computer Architecture, Real-Time Communication/Multimedia Stream

Work Experience

CUDA Engineer - University of Wisconsin-Milwaukee **Milwaukee, USA | Jan 2024 – Present**

- Developed and optimized **CUDA** kernels for accelerating ML inference using custom memory hierarchies, warp-level primitives, and tensor core utilization on **NVIDIA H100** GPUs.
- Leveraged **Nsight Systems** and **Nsight Compute** to analyze GPU occupancy, memory throughput, instruction bottlenecks, and warp divergence for performance tuning.
- Integrated **CUDA**-optimized kernels into **PyTorch** inference flows and benchmarked performance using **TensorRT** for transformer-based workloads.
- Tuned mixed-precision execution (**FP16**/**BF16**) using **Tensor Cores** and **NVIDIA Apex** to improve latency and throughput under high batch-load inference.
- Implemented custom **kernel fusion** techniques to reduce memory overhead and kernel launch latency in real-time inference tasks.
- Optimized inference for the **Mistral** foundational model using both **vLLM** and native **PyTorch** transformer paths, injecting custom **CUDA** kernels into the forward pass.
- Applied **CUDA Graph** capture and **kernel fusion** techniques to reduce token generation latency across dynamic input shapes, significantly improving throughput consistency.
- Performed warp-level scheduling and memory tuning for attention and MLP blocks to minimize inference overhead in long-context **LLM** scenarios.

Skills: CUDA, C++, PyTorch, TensorRT, Nsight Systems, Nsight Compute, cuBLAS, NCCL, Apex, Mixed Precision, Multi-GPU Inference, Kernel Fusion.

Software Engineer - Capri Global Capital **Gurgaon, India | Sep 2022 – July 2023**

- Implemented a **Dead Letter queue-based Retry mechanism** to reduce the failure rate of Bulk API Calls by **30%**.
- Implemented Redis-based cache mechanism in all the backend services to reduce the response time to an average of **150ms**.
- Developed a comprehensive Loan Distribution Platform from scratch, writing CSS manually without a framework, and integrated it with multiple APIs.
- Managed end-to-end deployment of the platform, ensuring smooth and reliable operation.
- Created a dynamic, client-side statement widget that transformed data into a PDF format, enhancing user accessibility and document generation.

Skills: Microservices based Back-End API Development,AWS (all popular services),CI/CD pipelines, Apache Kafka, Python, Django, MySQL, MongoDB, Redis, Test Driven Development, Distributed System, React, Redux, React Native, HTML, CSS, TypeScript.

Software Engineer - Udaan **Bangalore, India | June 2021 – Sep 2022**

- Boosted onboarding rate **24%** by developing a portal allowing sellers to bulk-upload product listings via Excel sheets, streamlining the process and reducing warehouse idle inventory.
- Addressed duplicate SKU creation issue in Udaan's warehouse system caused by concurrent product inwards. Implemented Redis-based SKU lookup with a **TTL greater than Cosmos DB replication time** to ensure data consistency and prevent duplicate SKUs.
- Enhanced the official e-commerce catalog page, optimizing user experience and interface design.
- Authored reusable unit tests using Jest and React Testing Library (RTL), achieving over **80%** test coverage, significantly improving code quality and early bug detection.

Skills: Microservices Back-End API Development, Cosmos DB, Java, DropWizard, Kotlin, Redis, Test Driven Development, Distributed Systems,CI/CD pipelines, Kubernetes, Docker, React, Redux, React Native, HTML, CSS, TypeScript.

Software Engineer - PhonePe **Bangalore, India | Feb 2020 — June 2021**

- Created a tool for reordering the gift card on the basis of user interest with drag and drop, which increased the gift card conversion rate by **10%**.
- Developed a **dashboard builder framework** using the concept of microfrontend to streamline the creation of new dashboards quickly and efficiently. It includes a comprehensive permission management system to control user access levels and a common authorization mechanism to simplify authentication and access control.

Skills: Microservices Back-End API Development, NodeJS, Java, ReactJS, Distributed Systems, Docker, React, Redux, HTML, CSS.

Software Engineer - Deepflux **Noida, India | April 2019 — Jan 2020**

- Automated mail report through cron jobs whenever an anomaly is found in datasets using luminoil (a LinkedIn library).
- Built monolithic backend services from scratch.
- Built a sophisticated analytics dashboard using Highcharts.js and React, performing seamless API integration for real-time data visualization.
- Automated front-end deployment processes, enhancing efficiency and reducing deployment errors.

Skills: Back-End API Development, HighChartJS, Flask, Python, Supervisor, PostgreSQL, React, TypeScript, HTML, CSS.

Key Achievements

- **2nd Place - QBI 2025 Hackathon at UCSF** : Secured 2nd place at the QBI 2025 Hackathon hosted by the University of California, San Francisco. Collaborated with researchers from UC Berkeley and UCSF to build **DECODE**, a transformer-based tool for predicting drug-protein binding efficacy. Used **Meta's ESM** and a Molecular Transformer to achieve 80% test accuracy on BindingDB, and developed a GUI for large-scale drug repurposing. **Feb 2025**

Research Projects

GPU Performance Optimization **Aug 2023 – Dec 2023**

- Designed and benchmarked custom CUDA kernels to enhance deep learning pipeline performance.
- Applied warp-level parallelism to minimize idle threads and maximize SM utilization.
- Implemented mixed-precision execution (FP16/BF16) to accelerate inference and reduce memory footprint.
- Optimized memory access with coalescing to reduce global memory latency.

ResNet Acceleration on CIFAR-10 **Jan 2024 – Mar 2024**

- Developed a CUDA-optimized image classification pipeline using warp primitives and memory coalescing.
- Replaced PyTorch convolution, ReLU, and SoftMax layers with custom CUDA kernels for fine-grained GPU control.
- Benchmarked training and inference performance vs. standard PyTorch baseline.

Custom CUDA Kernels for Softmax and Matrix Ops **Apr 2024 – Jun 2024**

- Implemented SoftMax and matrix multiplication kernels using warp-level primitives and coalesced memory.
- Measured GPU occupancy and throughput using Nsight Compute, achieving 35% improvement over PyTorch native ops.
- Reduced kernel launch latency and improved arithmetic intensity in classification layers.

TensorRT Inference Optimization with FP16 **Jul 2024 – Sep 2024**

- Converted ResNet18/34 models to TensorRT and applied FP16 mixed-precision optimization via Tensor Cores.
- Reduced batch inference latency by 43% using kernel fusion and optimized memory transfers.
- Validated model accuracy stayed within 1% of original post-quantization.